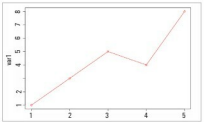

Figure 6: Plotting with R

Figure 7: Logo of Isabelle

```
In [1]: from scipy.special import factorial

In [2]: factorial(100, exact=True)
Out[2]: 93326215443944152681699238856266700490715968264381621468592963895217599993229915
608941463976156518286253697920827223758251185210916864000000000000000000000000L
```

Figure 8: Factorial using SciPy

software licensed under the GNU General Public License. It is often used as an open source alternative to proprietary statistical packages like SAS, SPSS, etc. But R being freely available is not the main reason for its widespread adoption by the statistical community; rather, its popularity is based on its power. R can be used as a command line utility as well as with a graphical user interface (GUI). A very popular GUI for R is called the RStudio. R is also capable of performing various graphical techniques. Figure 6 shows the output of the simple R commands 'var1 <- c(1, 3, 5, 4, 8)' and 'plot(var1, type="o", col="red")' when executed on the console.

Isabelle

While choosing ten useful mathematical tools for this article, we made sure that what we selected is used in diverse fields of mathematics, rather than choosing ten popular software from a single field, like numerical computations, for instance. So, this tool may not be as popular as many other mathematical this tools left out from this list. Let us clarify further—the general public is familiar with numerical computing tools like MATLAB, Scilab, etc, but may not be aware of SageMath or Maxima because the latter two are mostly used in the more abstract areas of mathematics like combinatorics, graph theory, number theory, etc. Mathematics itself is often considered an abstract subject – sometimes even by famous mathematicians. G.H. Hardy once said, "We have concluded that the trivial mathematics is, on the whole, useful, and that the real mathematics, on the whole, is not." There are certain fields in mathematics which are deemed abstract even by the standards of professional mathematicians; one such field is logic and automatic theorem-proving. Not many people are interested in automatic-theorem proving, but for the select few who work in this area, a tool to aid their quest will be like a gift from heaven. So, with that rationale, we introduce Isabelle, an interactive theorem-proving software. It is free, open source and available for use under the BSD licence. Isabelle has been developed by using two programming languages, ML and Scala. It can be used to encode first-order logic, higher-order logic, etc, for further processing. Figure 7 shows the logo of Isabelle.

SciPy

SciPy is an open source Python library used for scientific computing. It contains modules for optimisation, linear algebra, calculus, interpolation, Fast Fourier Transforms, signal processing, image processing, etc. SciPy is part of the NumPy stack, which also includes tools like Matplotlib, SymPy, etc. Matplotlib is a plotting library and SymPy is a library for symbolic computing in Python. The SciPy library is available under the BSD licence. Figure 8 shows how SciPy is used to find the factorial of 100. Even in a modest computer system, we were able to find the factorial of 100,000. But we will not dare to show the output in this article because the resulting number is 456574 digits long. The December 2017 issue of *OSFY* had 108 pages, including the covers. The page we randomly chose in that issue had 54 lines of text and the line we randomly chose had 123 characters, so, a simple calculation will tell you that it will take at least 68 pages of the next issue of *OSFY* to print this number. This tells us about the power of SciPy. And with a powerful computer, you could do wonders with SciPy.

gnuplot

Scientific research involves representing data in a neat and concise manner. Graphs are often used to do this. It's for this purpose that we can use gnuplot, a command line program that can plot 2D and 3D graphs of functions and data. It is frequently used in vector graphics. gnuplot can produce output in many different image formats like Portable Network Graphics (PNG), Encapsulated PostScript (EPS), Scalable Vector Graphics (SVG), Joint Photographic Experts Group (JPEG), etc. It is a cross-platform utility that runs on a variety of operating systems like UNIX, Linux, Microsoft Windows, MacOS, etc. Even though the name of the software starts with 'gnu', it is not part of the GNU project. In fact, the inclusion of gnuplot in this list can be contested because, in a very strict sense, it is not free and open source software because it is licensed under the gnuplot licence, which gives users the right to modify the source code but the right to distribute modified versions is withheld. But the extreme popularity of gnuplot made us select it rather than a free and open source graph plotting utility like xgraph. Figure 9 shows a 3D graph plotted using gnuplot.

LaTeX

Finally, let us think about the life of a brilliant mathematician who has done a lot of research –with and without using some of the mathematical tools we have listed just now—to come up with some excellent theorems. It's eventually time to write down these results

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